

Satyam Kushwah

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Education

Indian Institute of Technology (ISM), Dhanbad

Bachelor of Technology in Chemical Engineering

Aug 2021 - May, 2025

Dhanbad, Jharkhand

Experience

American Express

Sept 2025 – May 2026

Data Science - Credit Risk

- Engineered an **automated case review tool and vendor model tracking tool** for US and international markets, **reducing manual effort by over 50%** and supporting analysis of high-volume cases monthly.
- Reviewed multiple **high-balance (\$30Mn+)** risk cases and delivered actionable insights to **enhance model interpretability using SHAP-based explanations, Bureau Information, and transaction-level analysis.**
- Identified early model drift across Gini, accuracy, and stability metrics; proposed targeted recalibration strategies contributing to a 5% performance improvement.
- Led CDIT tracking using **10-day rolling averages across max, min, mean, median, and missing% metrics (0% threshold);** generated alerts on control limit breaches and **performed RCA to determine acceptance or rejection.**

mCaffeine

May 2025 – Aug 2025

Data Analyst and Automation Intern

- Created a real-time Power BI dashboard analysing **900K+ orders across multiple channels, strengthening regional utilisation tracking** and enabling faster, data-driven operational and strategic decisions.
- Improved delivery efficiency by **4.6% by identifying regional performance gaps, benchmarking logistics KPIs, and recommending targeted process optimisations** to enhance turnaround time and resource allocation.
- Designed and deployed an internal order fulfillment dashboard, eliminating **70+% manual reporting overhead through end-to-end data integration, transformation, and real-time visualization pipelines.**
- Built a master inventory framework with **15+ structured attributes, centralising inventory planning processes** and reducing product listing errors by **30%, improving catalogue accuracy and supply visibility.**

Projects

Financial anomaly detection with LLM Integration

- Built a real-time fraud detection system on 389K transactions integrating ML and LLM-powered case narration
- Trained Random Forest and Isolation Forest on 23 features (Job of Card Holder, Merchant Risk, Distance, Declines); integrated GPT-4o-mini pipeline to auto-generate structured analyst case summaries for every flagged transaction.
- Achieved 97% accuracy and 0.98 weighted F1 across 389K transactions; documented precision-recall tradeoff on a 97:3 imbalanced dataset, proposing threshold tuning and SMOTE to improve fraud-class recall beyond 15%.

Achievements

Cleared JEE Mains Examination 2021 & JEE Advanced Examination 2021 by **being in the top 1%.**

Achieved a **rank in top 50 in the Meesho Dice** competition in the Software domain.

Ranked in the **top 0.01% in the Walmart SparkPlug** competition, earning an internship opportunity with Walmart.

Excelled in the Doramon Den Boot camp for Product Management, demonstrating outstanding performance

Secured a **rank of 20 out of 500** entries in Kaggle KDSH Data Solution Competition.

Granted **MMVY Scholarship (top 5%)** for excellent performance in Class 12th, JEE Mains Advanced 2021 by MP Government

Technical Skills

Programming: Python, SQL, SAS, C, C++

Data Analysis & ML: Pandas, NumPy, Scikit-learn, TensorFlow

Visualization: Power BI, Tableau, Excel

Other Tools: MATLAB, HTML, CSS, JavaScript

Social Engagements

Sponsorship Team, Organizer of Srijan 23', the Annual Culture Fest of IIT(ISM) Dhanbad

Design Team, Head of Turbulence 23', Annual Chemical Engineering Symposium

Graphic Designer Head of Chemical Engineering Society, IIT (ISM) Dhanbad- Designing and Tech Team

Club Member of Lights Camera ISM Club- Videography, Sound Designing, Video Editing